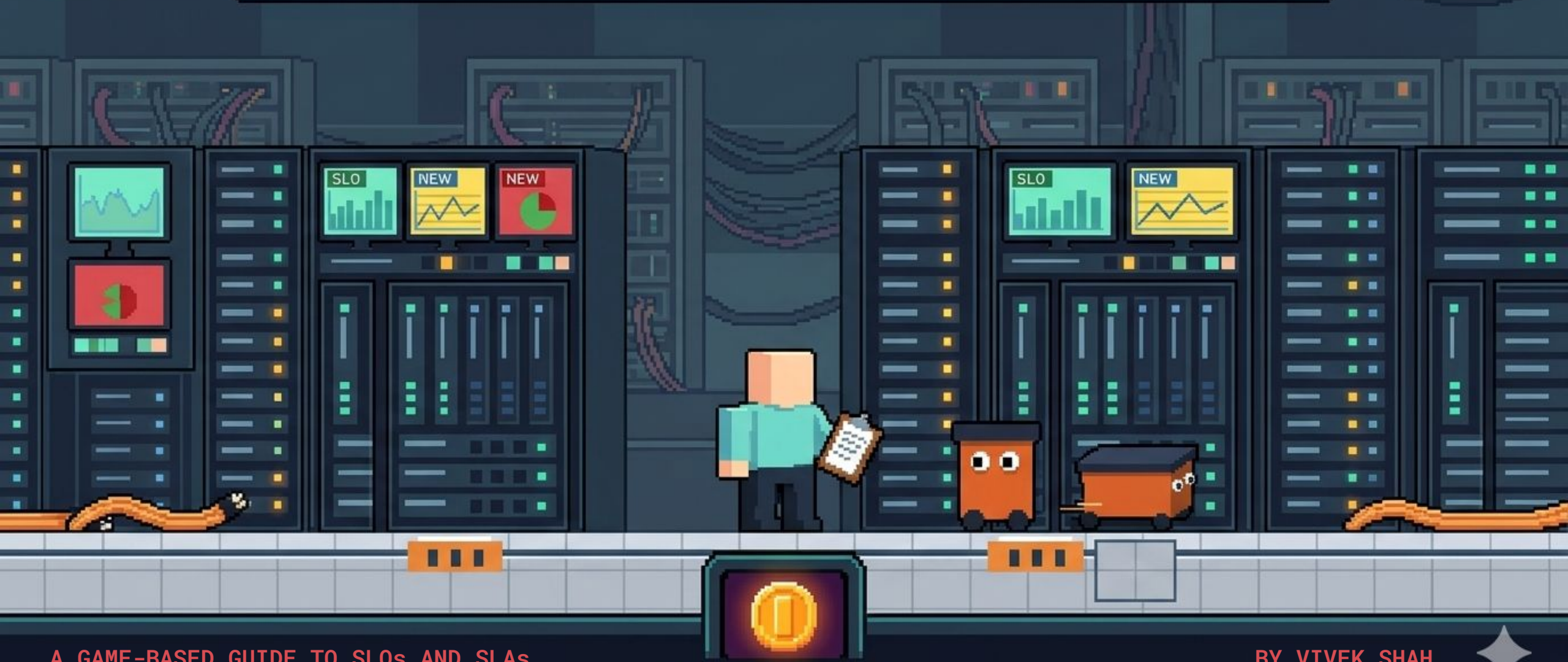


99.9% FUN



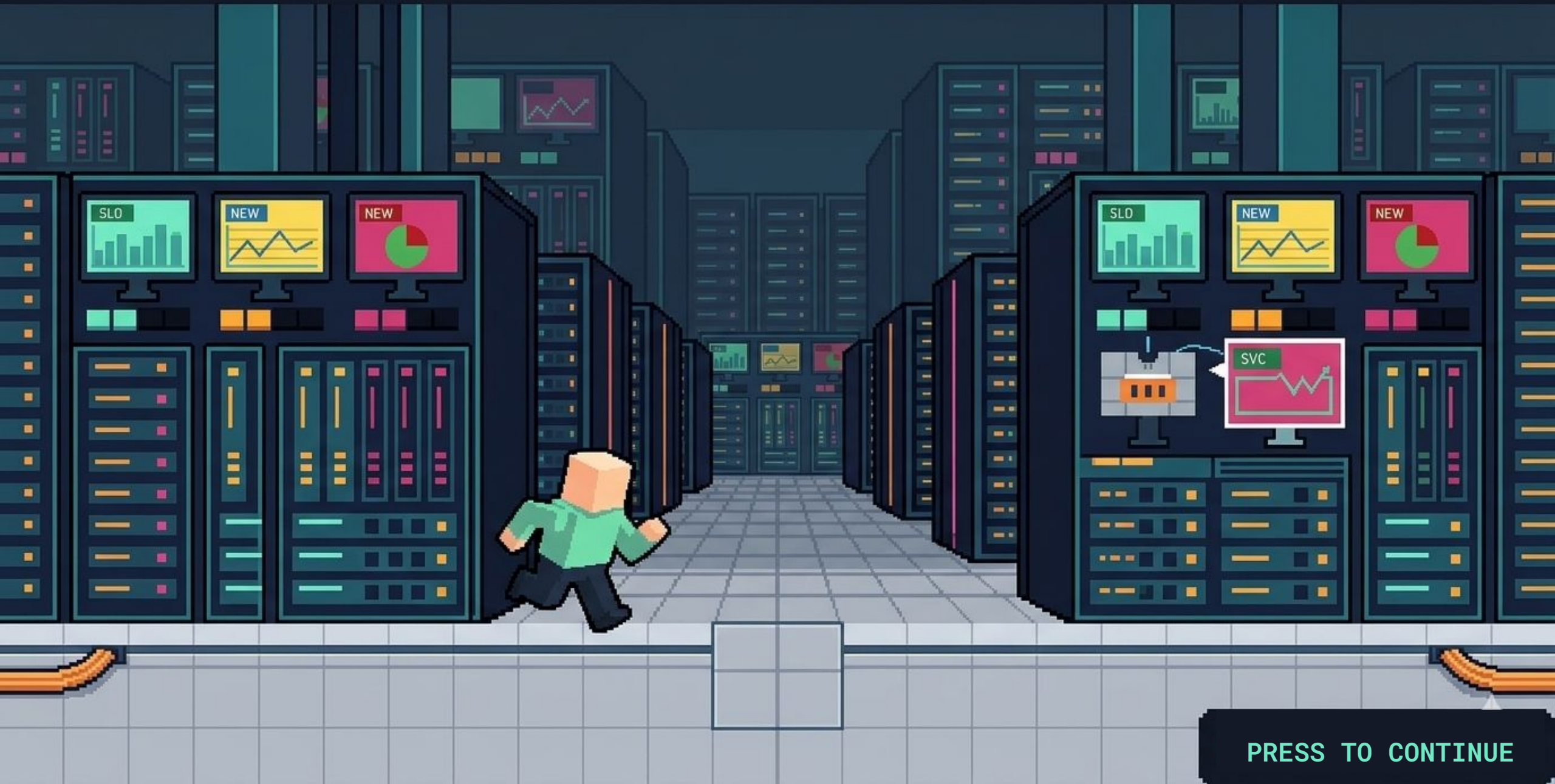
A GAME-BASED GUIDE TO SLOs AND SLAs

INSERT COIN

BY VIVEK SHAH



Reliability is a game you're already playing.



PRESS TO CONTINUE

LEVEL 1: TUTORIAL.



SLI × SLO × SLA



Gauge × Line × Lawyer



SLI

What you measure

The speedometer

Ratio of good
events to valid
events

SLO

What you promise
yourself

"Don't go above
70"

Target on that
ratio

SLA

What you promise
the world

The ticket the cop
writes

Contract with
consequences

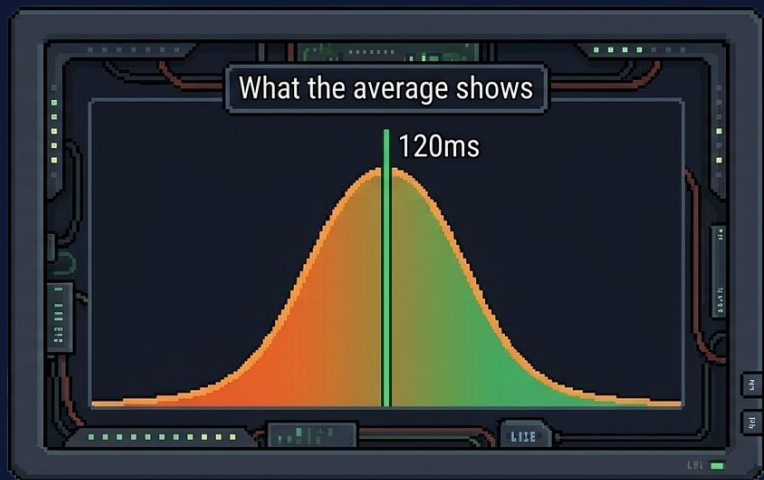


LEVEL 2: CABLE HAZARDS.

Latency & Percentiles

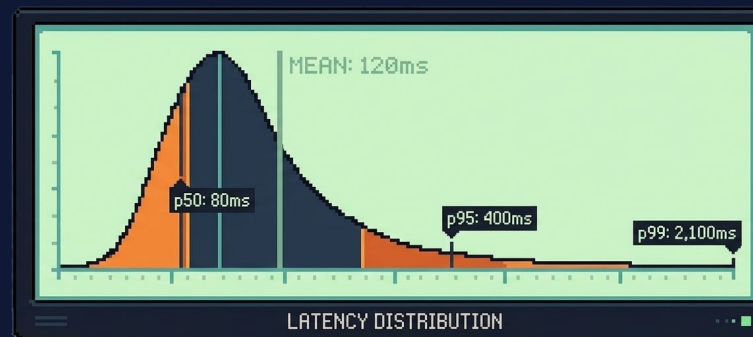


AVG: 120ms — FINE.



Same average.

p99: 2,100ms — TICKET FILED.



Very different
users.

1-STAR
REVIEW



The same SLI tells different stories at different zooms.

5-minute window



catches incidents

1-hour window



catches
degradations

30-day window



catches trends &
budgets



LEVEL 3: BUGGY CODE



Error Rate, Error Budgets, Burn Rate

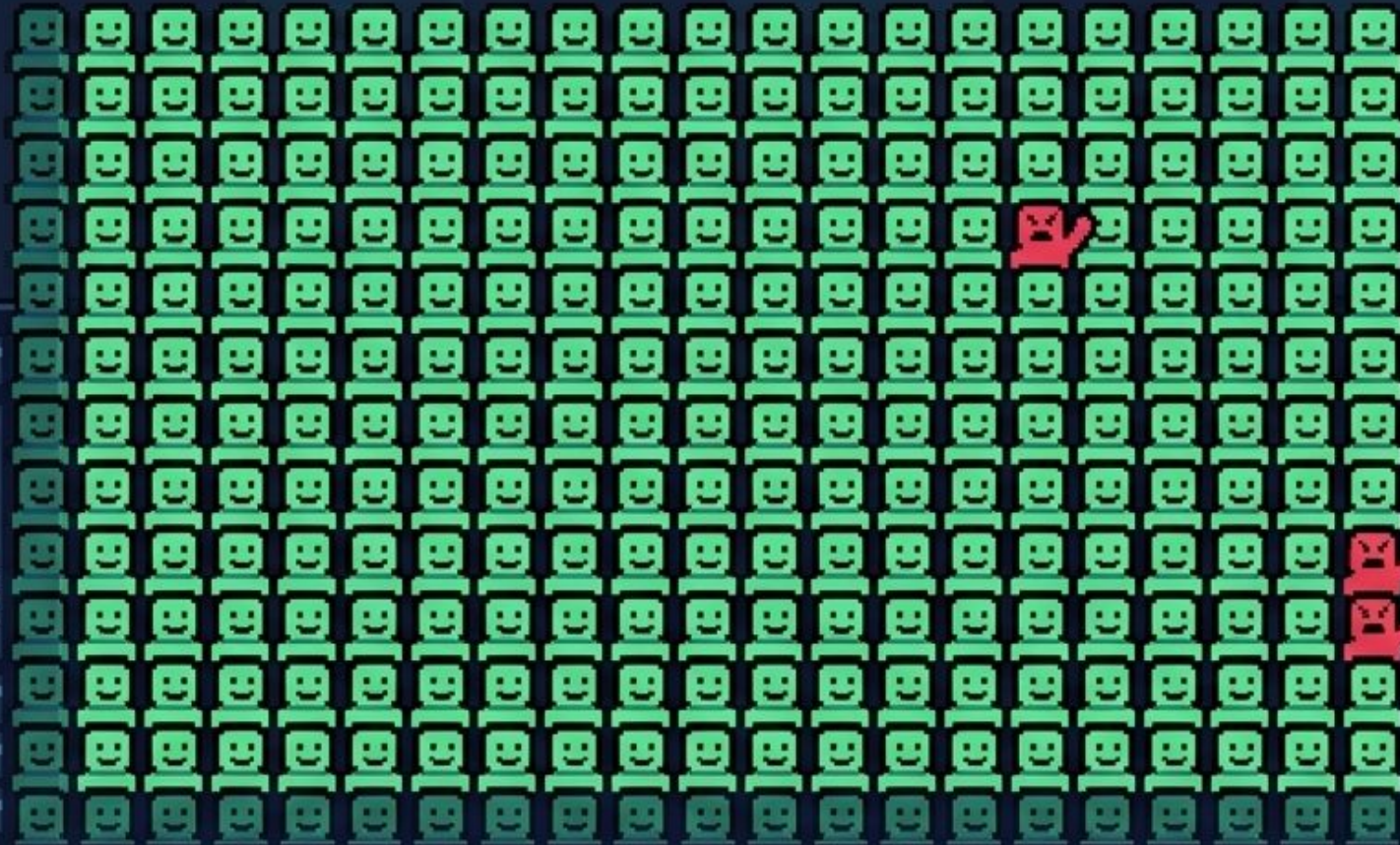


SCORE: 159 STATUS
SCORE: 



99.7% success. 3 unhappy users. Out of how many?

Error rate = bad events ÷ valid events



99.9% over 30 days = ~43 minutes 12 seconds of "allowed wrong"

ERROR BUDGET



43:12 REMAINING

YOU DON'T HAVE A TARGET.



YOU HAVE A BUDGET.



FROM THE FIELD: PROVIDENCE HEALTH

MESSAGES API

Window: 30-day rolling



ERROR RATE

SLA: $< 0.5\%$
SLO: $< 0.25\%$



side-quest



SLO
SLA
LATENCY



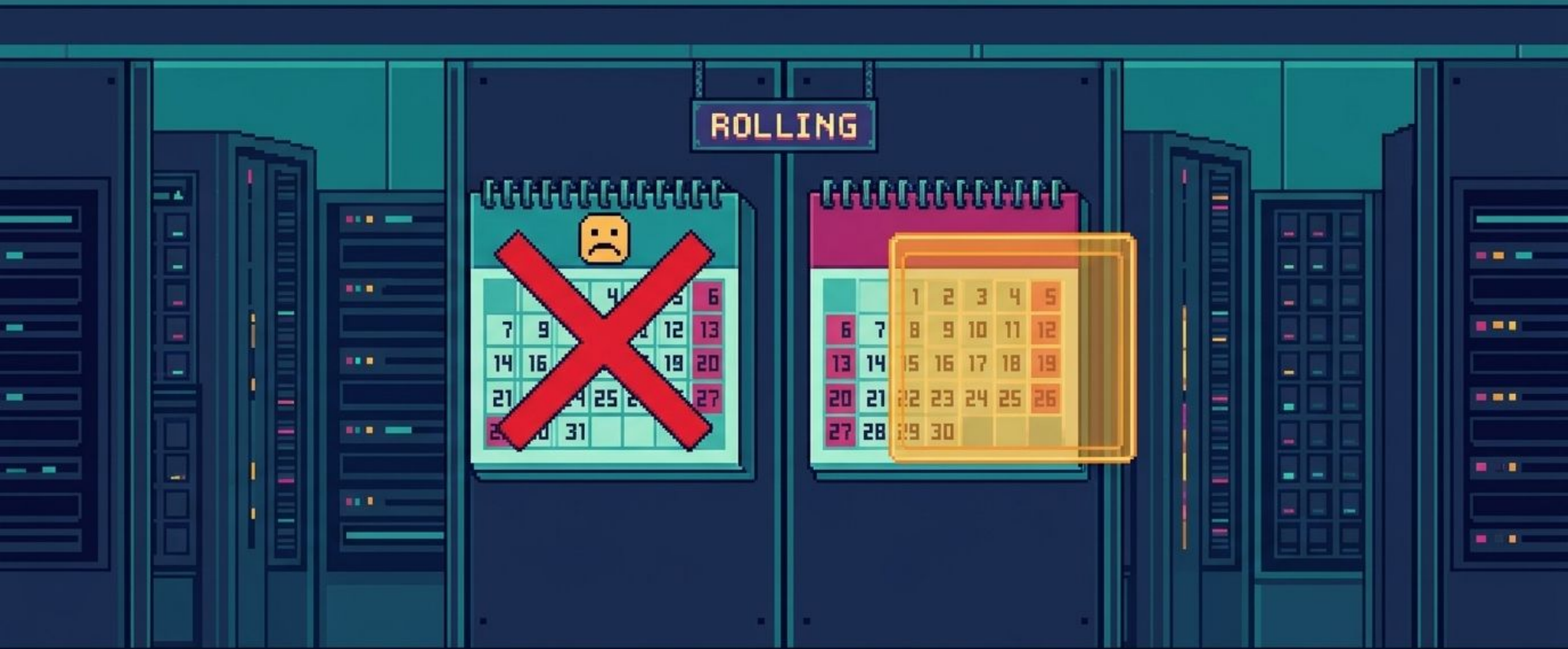
SLO
SLA
ERROR RATE



SLO
SLA
UPTIME

When a patient can't message their provider,
the budget isn't a metric. It's a decision.

99.9% over 30 days = ~43 minutes 12 seconds of "allowed wrong"



The budget heals continuously. No free Fridays.





Sustained 6h



2x

~1.7% Budget

Sustained 20m



14x

~6.5% Budget

Sustained 90s



50x

~0.2% Budget

BURN RATE = how fast you're
spending budget vs. normal.
1x = on schedule.



Which one pages
you at 3 AM?

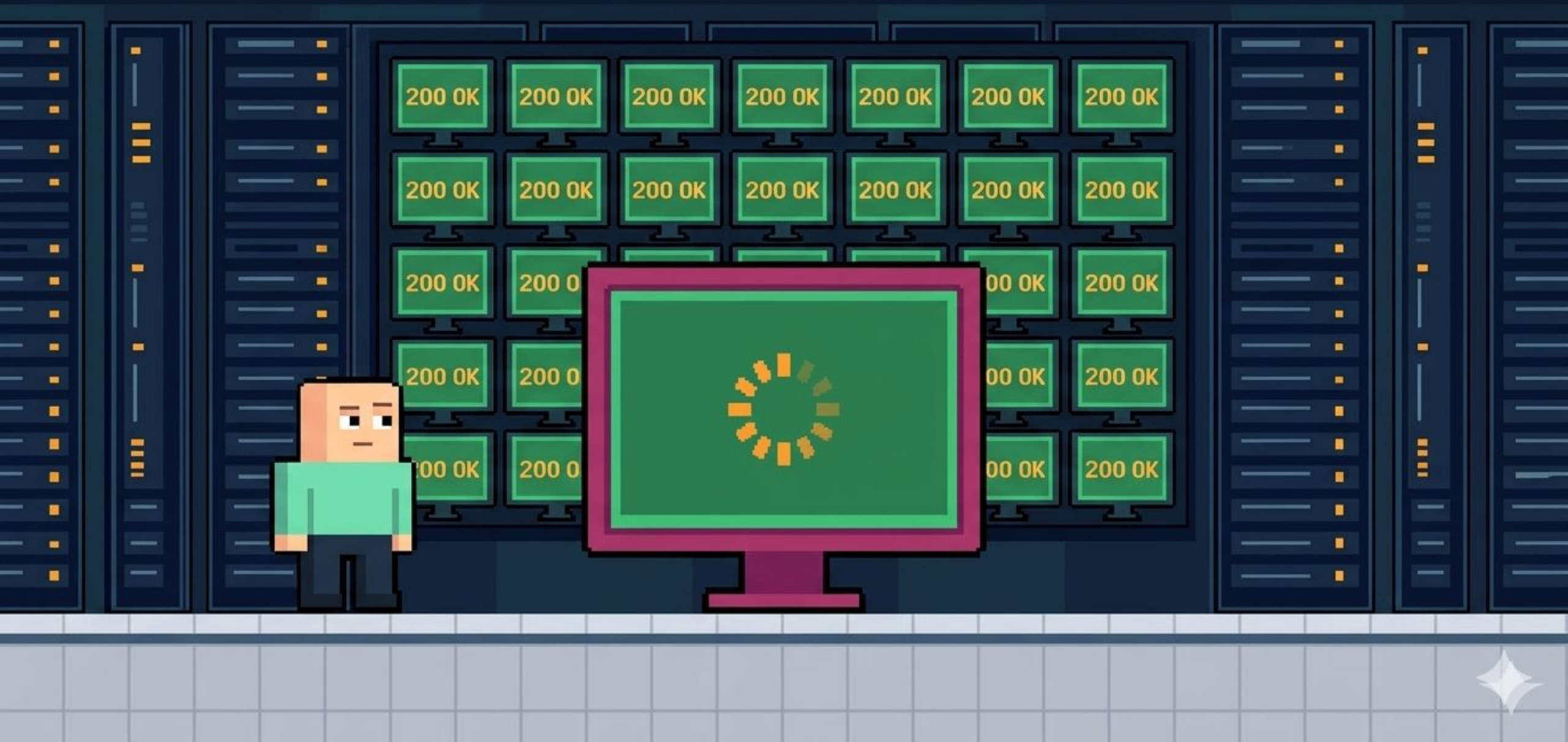
VOTE

Availability × Uptime

LEVEL 4: THE UNPLUGGED MONITOR



Uptime is a heartbeat. Availability is a conversation.





LEVEL 5: BOSS FIGHT

AI as a Subject of SL0s



CONFIDENTLY
WRONG



The CEO of Acme Corp
is Jane Doe.

(source: acme.com/about, 2019)

Jane Doe does not exist.
acme.com/about returns 404.

I'M DOING GREAT!
EVERYTHING
IS OKAY!

LATENCY
OK

ERROR RATE
0%

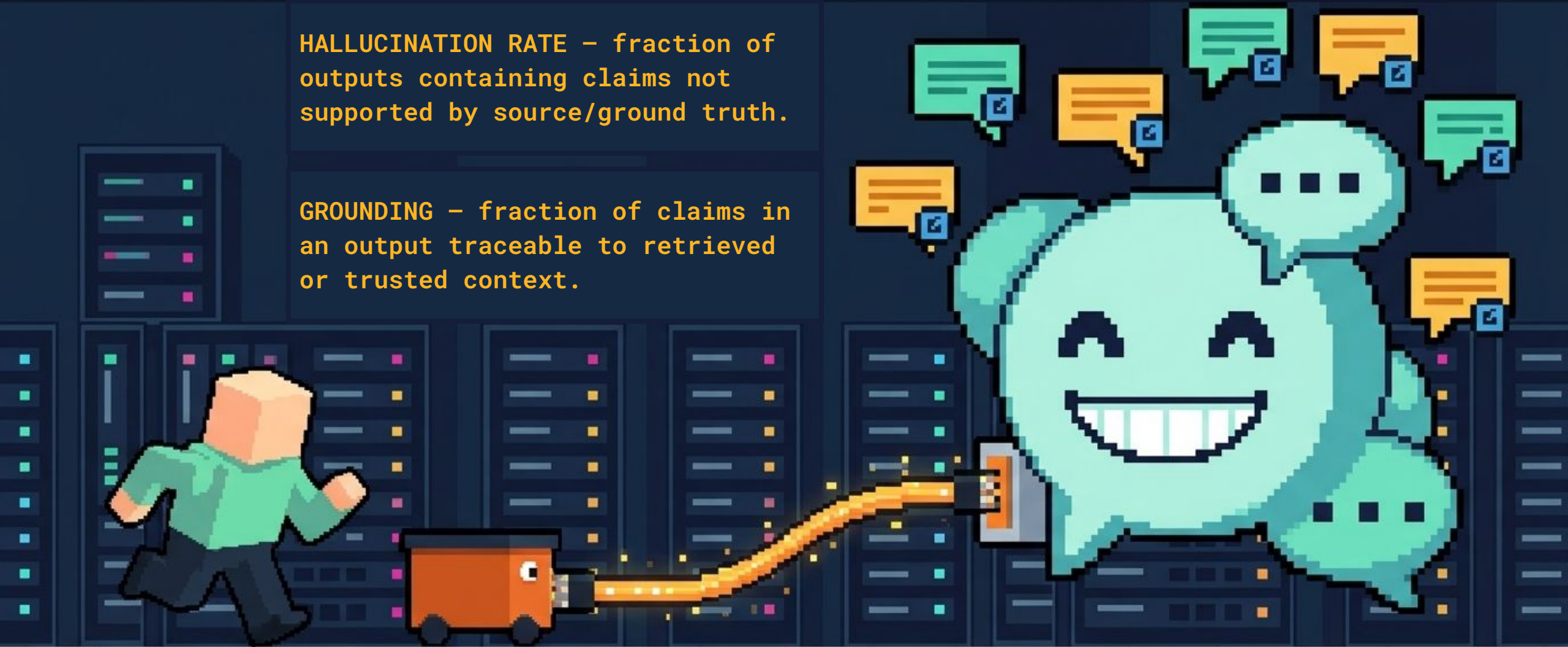
UPTIME
100%

GROUNDING: FAIL

GROUNDING

HALLUCINATION RATE – fraction of outputs containing claims not supported by source/ground truth.

GROUNDING – fraction of claims in an output traceable to retrieved or trusted context.



Measurement is hard. Sampling + LLM-as-judge + human review. That's the honest stack. ✨

SELECT YOUR SLI.



Availability



Latency to First token



Latency



Context Utilization



Error Budget



Hallucination Rate



Uptime



Grounding

If your AI dashboard only has latency and error rate on it

you've built a thermometer for a system whose problem is lying.



The model is only as smart as the briefing you gave it.

service purpose → architecture → observability stack → historical incidents
dependency guarantees → user sensitivity → org context → regulatory constraints



- $SLI \times SLO \times SLA$
- Average $\times$ Percentile Latency
- Error Rate $\times$ Error budget
- Availability $\times$ Uptime
- Traditional $\times$ AI-era SLIs



1. Your SLO is the line. Your SLA is the lawyer.
2. The average user does not exist. The p99 user files tickets.
3. Threshold thinking asks did it bleed? Burn-rate thinking asks will it bleed out?
4. Uptime is a heartbeat. Availability is a conversation.
5. A 200 OK can still be a lie.



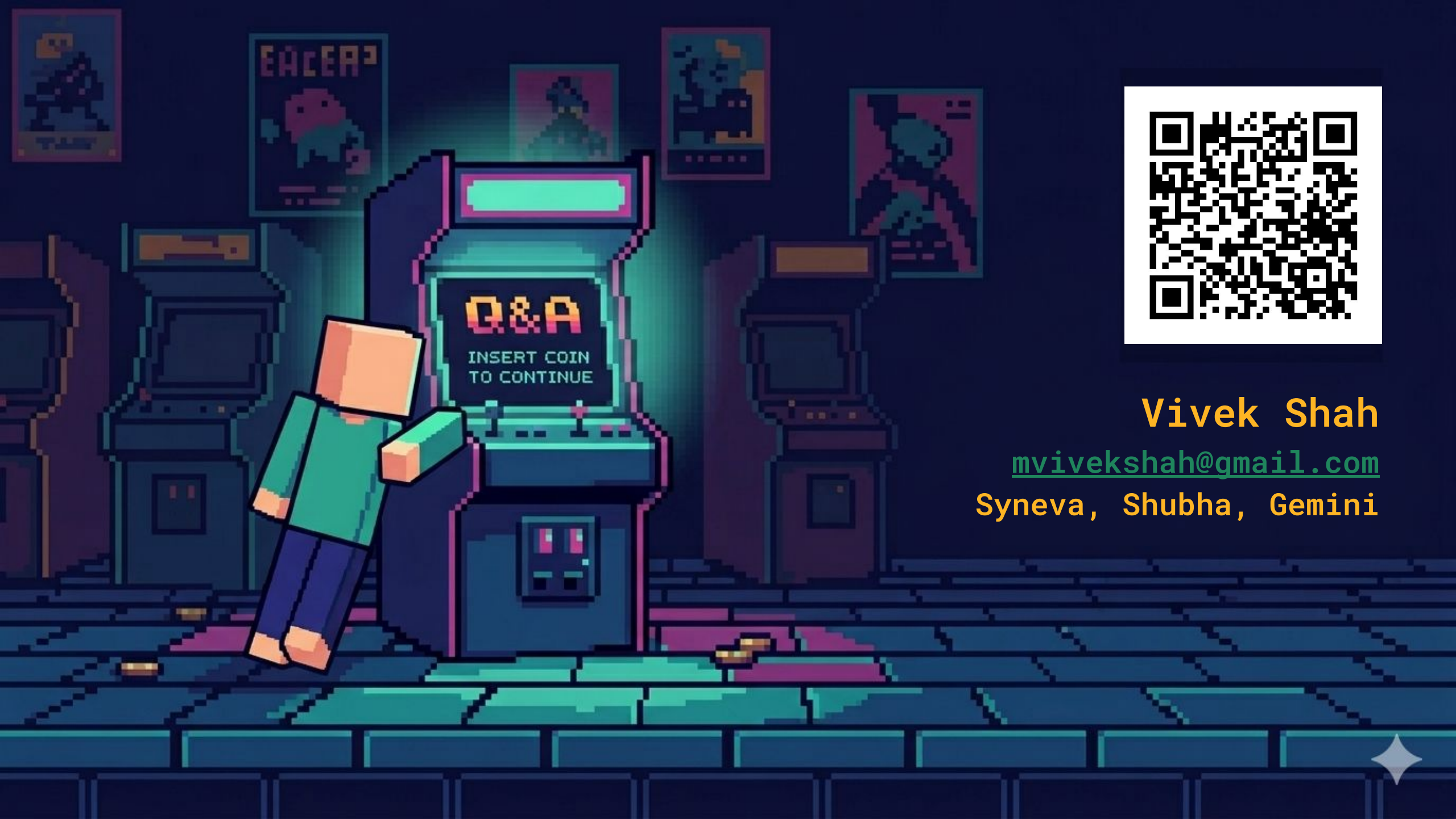
GAME SAVED



SLIs are what you see.
SLOs are what you promise yourself.
SLAs are what you promise the world.
Error budgets are how
you stay honest about all three.

And in the AI level...
being up is not the same as being right.





Vivek Shah

mvivekshah@gmail.com

Syneva, Shubha, Gemini

Additional Thoughts

- Evernote
- How do you use SLOs or SLAs